Hanna Ayalew

BLESSED G/MICHAEL CATHOLIC SECONDARY SCHOOL

PHYSICS MODEL EXAM FOR GRADE-12 **GINBOT 2018 / JUNE 2026**

NUMBER OF ITEMS: 60 **TIME ALLOWED: 2:30 HOURS**

GENERAL DIRECTIONS:

Each of the following questions is followed by four possible alternatives. Read each question and carefully **write** only the **CAPITAL** letter of your choice on the separate answer sheet provided.

Any form of **cheating** or an attempt to cheat in the examination will result in an automatic **dismissal** from the examination hall and cancellation of your score (s).

You may refer to the information given below when you work on some of the questions.

CONSTANT	SYMBOL	VALUE
Acceleration due to gravity	g	$10\ m/s^2$
Elementary charge	q	$1.6 \times 10^{-19}\ C$
Permitivity of vacuum	ε_o	$8.85 \times 10^{-12}\ Nm^2/C^2$
Coulomb's constant	K	$9 \times 10^9\ Nm^2/C^2$
Permeability of free space	μ_o	$4\pi \times 10^{-7}\ F/m$
Density of water	ρ	$1000\ kg/m^3$
Speed of light in vacuum	c	$3 \times 10^8\ m/s$
$Sin30^0=cos60^0=0.5$ $sin60^0=cos30^0=0.87$ $sin37^0=cos53^0=0.6$	$sin53^0=cos37^0=0.8$ $sin45^0=cos45^0=0.707=\frac{1}{\sqrt{2}}$ and $\pi=3.14$	

1. Which of the following lists contains only fundamental physical quantities in the International System of Units (SI)?

A. Length, mass, electric current

C. Weight, volume, temperature

B. Force, acceleration, energy

D. Velocity, mass, time

2. A student measures the length of a copper rod using a vernier caliper with a least count of 0.01 cm. If the true length of the rod is 12.50 cm and the student records measurements of 12.31 cm, 12.32 cm, and 12.31 cm, how should these measurements be described?

A. High precision but low accuracy

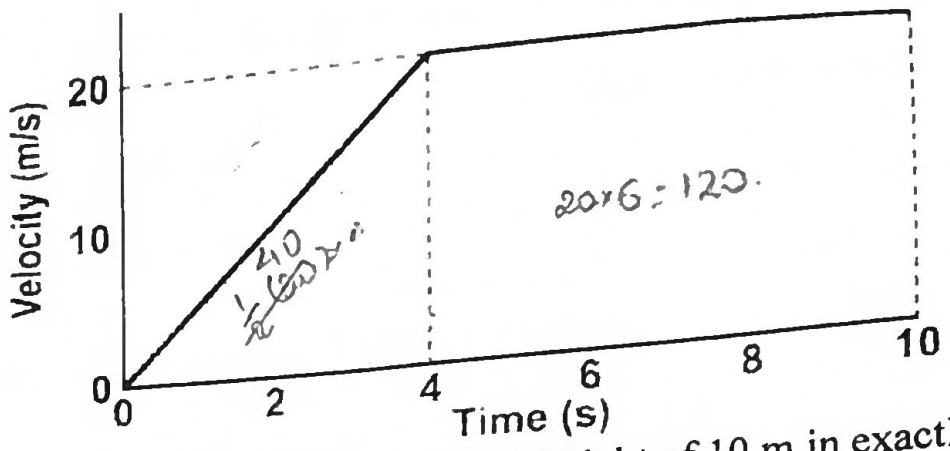
C. High precision and high accuracy

B. Low precision but high accuracy

D. Low precision and low accuracy

PHYSICS MODEL EXAM

3. The velocity-time ($v - t$) graph of a motorcycle traveling along a straight road in Hawassa is shown below. What is the total displacement of the motorcycle during this entire 10second interval?



A. 120 m

B. 160m

C. 200 m

D. 240 m

4. An electric motor lifts a crate weighing 600 N vertically upward through a height of 10 m in exactly 5 s at a constant speed. What is the power output delivered by the motor?

A. 6,000 W B. 1,200 W C. 30,000 W D. 300 W

5. A toy car accelerates uniformly from rest along a straight path. If it covers a distance of 20 m in the first 4 s, what distance will it cover during the next 4 s interval?

A. 40 m B. 60 m C. 80 m D. 20 m

6. A 2 kg block is dropped from rest from the top of a building. Just before hitting the ground, its kinetic energy is measured to be 360 J. If mechanical energy is conserved, what is the height of the building?

A. 9 m B. 36 m C. 18 m D. 72 m

7. The temperature of a solution in a laboratory experiment decreases by 25°C. What is the equivalent temperature change on the Kelvin scale?

A. 298 K B. 45 K C. 25 K D. -248 K

8. According to Newton's first law of motion, what happens to a moving object if the net external force acting on it suddenly drops to zero?

A. It comes to an immediate stop.

B. It turns in a circular path due to momentum.

C. It begins to gradually decelerate until it rests.

D. It continues moving along a straight path with constant speed.

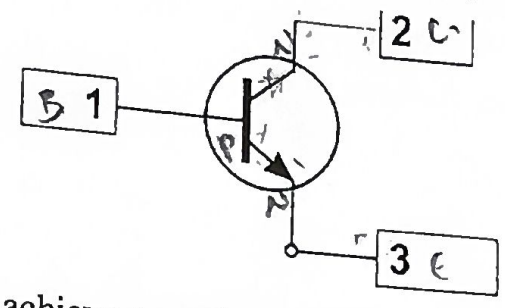
9. A simple inclined plane has a length of 5 m and a vertical height of 2 m. If a worker requires an effort force of 100 N to push a 200 N box up this ramp, what is the efficiency of the machine?

A. 40% B. 100% C. 50% D. 80%

10. A car travelling at 20 m/s brakes uniformly to a complete stop over a distance of 40 m. What is the magnitude of the car's deceleration?

A. 20.0 m/s^2 B. 2.5 m/s^2 C. 10.0 m/s^2 D. 5.0 m/s^2

11. An electronics device (Transistor) symbol is shown below a three-terminal. Identify the terminals and device type.



A. 1 = Emitter, 2 = Base, 3 = Collector; NPN
B. 1 = Collector, 2 = Base, 3 = Emitter; PNP
C. **1 = Base, 2 = Collector, 3 = Emitter; PNP**
D. 1 = Base, 2 = Emitter, 3 = Collector; NPN

12. Which of the following describes a machine that can achieve a mechanical advantage (MA) greater than 1?

A. A machine that increases the speed of displacement at the expense of force.
B. A machine that multiplies the magnitude of the effort force to lift a larger load.
C. A machine that changes only the direction of the force without changing its magnitude.
D. A machine that increases the required effort force.

13. Which characteristic of a sound wave determines its perceived pitch to a human listener?

A. Frequency
B. Waveform complexity
C. Amplitude
D. Wave speed

14. A sound wave traveling through air has a frequency of 680 Hz. If the speed of sound in air at that temperature is 340 m/s, what is the wavelength of this sound wave?

A. 0.5 m
B. 2,312 m
C. 2.0 m
D. 340 m

$\lambda = \frac{340}{680} = 0.5$

15. An open container holds a fluid of uniform density (ρ). According to the principles of fluid statics, how does the absolute hydrostatic pressure vary with depth below the surface?

A. It varies inversely with the square of the depth.
B. It increases linearly as depth increases.
C. It decreases linearly as depth increases.
D. It remains uniform at all depth coordinates.

16. An object moves with uniform acceleration along a straight path. Which parameter remains completely constant over time during this motion?

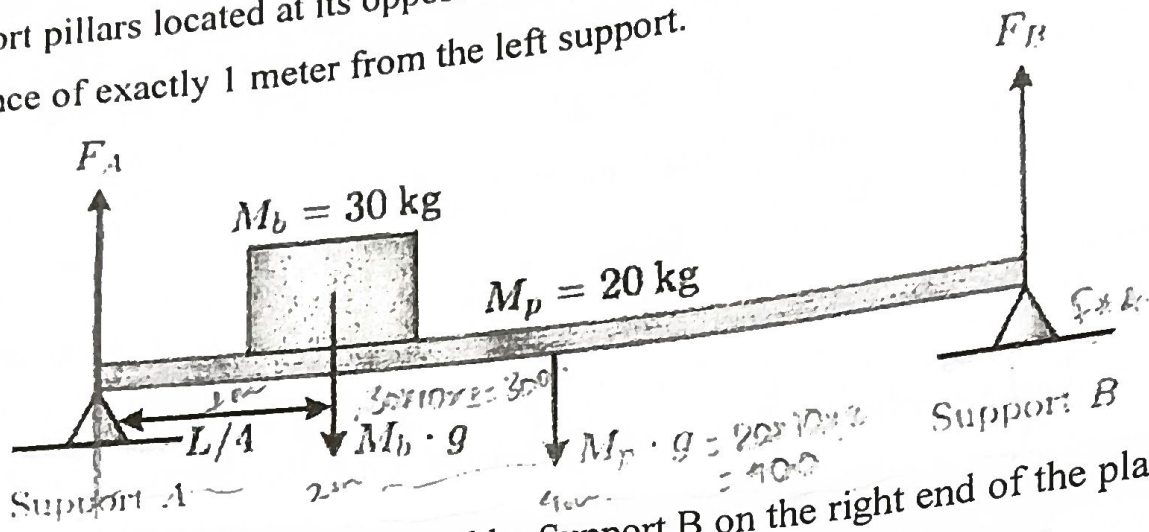
A. The velocity of the object.
B. The total distance covered per second.
C. The rate of change of velocity.
D. The displacement of the object.

17. A ray of light traveling inside air strikes a flat glass surface at an angle of incidence of 30° relative to the normal. If the refractive index of the glass is 1.50, what is the sine of the angle of refraction?

A. 0.75
B. 1.00
C. 0.50
D. 0.33

$n_1 = 1 \quad \sin\theta_1 = \frac{1}{2}$
$n_2 = 1.5 \quad \sin\theta_2 = ?$
$\frac{1}{1.5} = \frac{\sin\theta_2}{\frac{1}{2}}$
$\sin\theta_2 = \frac{1}{3} = 0.3$

B 18. A uniform horizontal wooden plank of mass 20 kg and total length 4 meters rests symmetrically on two sharp support pillars located at its opposite ends. A heavy block of mass 30 kg is placed on the plank at a distance of exactly 1 meter from the left support.



What is the upward normal force (F_B) exerted by Support B on the right end of the plank?

A. 125 N B. 175 N C. 250 N D. 325 N

C 19. Which of the following regions of the electromagnetic spectrum carries the highest photon energy and has the shortest wavelength?

A. Gamma rays C. Radio waves

B. Ultraviolet light D. Infrared radiation

A 20. An electrical circuit component has a resistance of 10 Ω. If a steady current of 3 A flows through this resistor for 20 s, how much electrical energy is dissipated as heat?

A. 1,800 J B. 600 J C. 60 J D. 3,600 J

C 21. Which of the following optical conditions is correctly matched with its correcting lens type?

A. Myopia (Nearsightedness) — Converging (Convex) Lens

B. Myopia (Nearsightedness) — Diverging (Concave) Lens

C. Hyperopia (Farsightedness) — Diverging (Concave) Lens

D. Astigmatism — Standard Spherical Convex Lens

A 22. An object is projected vertically upward from ground level with an initial velocity of 30 m/s. Neglecting air resistance, what is the maximum height reached by the object?

A. 45 m B. 30 m C. 90 m D. 150 m

C 23. An object is placed at a distance of 30 cm in front of a thin converging lens with a focal length of 10 cm. At what distance from the lens will the real image form?

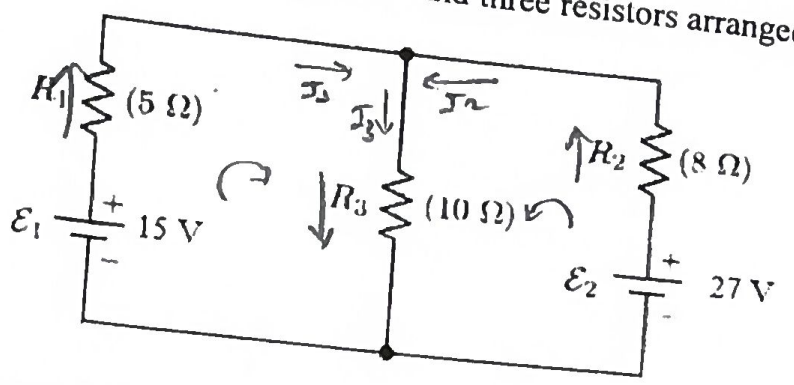
A. 40.0 cm B. 20.0 cm C. 15.0 cm D. 7.5 cm

D 24. Let **i** and **j** represent orthogonal unit vectors along the x and y axes. What is the value of the scalar dot product expression $(3\mathbf{i} + 2\mathbf{j})\cdot(2\mathbf{i} - 4\mathbf{j})$?

A. Zero B. 14 C. $6i-8j$ D. -2

Set by Abebaw Tilahun Contact: +251 921 279 547 or @abty01.me or abtytutorial@gmail.com

25. Consider a two-loop circuit network containing two ideal DC batteries and three resistors arranged as shown. What is the magnitude of the electric current flowing downward through the middle shared resistor (R_3)?



A. 1.0 A
B. 1.5 A
C. 2.0 A
D. 3.0 A

26. Two vectors **P** and **Q** lie in the xy-plane. Vector **P** points along the positive x-axis, and vector **Q** points along the positive y-axis. In which direction does their cross product vector **P** × **Q** point?

A. Along the positive y-axis
B. Into the origin point
C. Along the negative x-axis
D. Along the positive z-axis.

27. A stunt bike is ridden off a horizontal ledge at a speed of 15 m/s. If it hits the flat ground below exactly 2 s later, what is the horizontal distance from the base of the ledge to the landing point?

A. 20 m
B. 60 m
C. 45 m
D. 30 m

28. Which of the following states the main operating principle behind traditional liquid-in-glass thermometers?

A. Thermal conduction rates differ between liquids and glass containers.
B. The boiling point of the liquid changes depending on ambient pressure conditions.
C. Liquids experience measurable volumetric expansion when their temperature increases.
D. Glass shrinks significantly when exposed to thermal radiation.

29. A 5 kg block is placed on a rough horizontal table. The coefficient of static friction between the block and the table is 0.40. What horizontal force is required to just start moving the block from rest?

A. 20 N
B. 2 N
C. 12.5 N
D. 50 N

30. A small domestic radio receiver uses a compact battery pack that delivers a constant current of 25 mA continuously for exactly 3 hours. What is the total amount of electrical charge that flows through the radio circuit during this operating window?

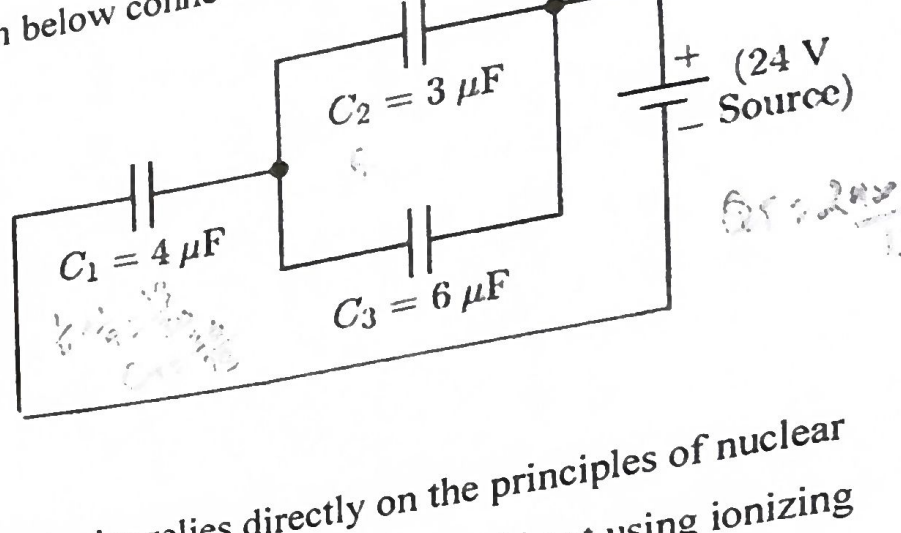
A. 75 C
B. 4500 C
C. 270 C
D) 0.075 C

31. What type of majority charge carrier is introduced into an intrinsic semiconductor crystal structure when it is doped with a pentavalent impurity element like phosphorus?

A. Free electrons
B. Positive ions trapped in the lattice
C. Neutrons
D. Positive holes

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32. Consider the active capacitor network diagram below connected across an ideal 24-V DC source. What is the magnitude of the electrostatic charge stored specifically on the plates of the C_1 capacitor?



A. 144 μC

B. 8.66 μC

C. 66.46 μC

D. 96 μC

33. Which of the following medical imaging technologies relies directly on the principles of nuclear magnetic resonance (NMR) to produce detailed internal structural images without using ionizing radiation?

A. Magnetic Resonance Imaging (MRI)
C. X-ray Radiography
B. Computed Tomography (CT) scan
D. Ultrasound Sonography

34. An ideal incompressible fluid flows through a horizontal pipe with a varying cross-sectional area. If the fluid's speed is 4 m/s in a section of the pipe with an area of 0.10 m^2, what is its speed in a section where the area narrows to 0.02 m^2?

A. 40.0 m/s
B. 20.0 m/s
C. 8.0 m/s
D. 0.8 m/s

35. An elevator cab of mass 1,000 kg is accelerated vertically upward at a rate of 2 m/s^2. What is the tension in the supporting cable?

A. 8,000 N
B. 2,000 N
C. 12,000 N
D. 10,000 N

36. A 3 kg rifle fires a bullet of mass 0.03 kg horizontally with a muzzle velocity of 400 m/s. What is the magnitude of the recoil velocity of the rifle?

A. 12.0 m/s
B. 2.0 m/s
C. 40.0 m/s
D. 4.0 m/s

37. A parallel-plate capacitor is charged by connecting it temporarily to a DC battery and is then disconnected from the power source. If the separation distance between the plates is now carefully pulled apart to twice its original width using insulated tools, what happens to the electric field (E) and the energy (U) stored in the capacitor?

A. E doubles, U stays the same.
C. E stays the same, U doubles.
B. E stays the same, U reduces to half.
D. Both E and U reduce to half.

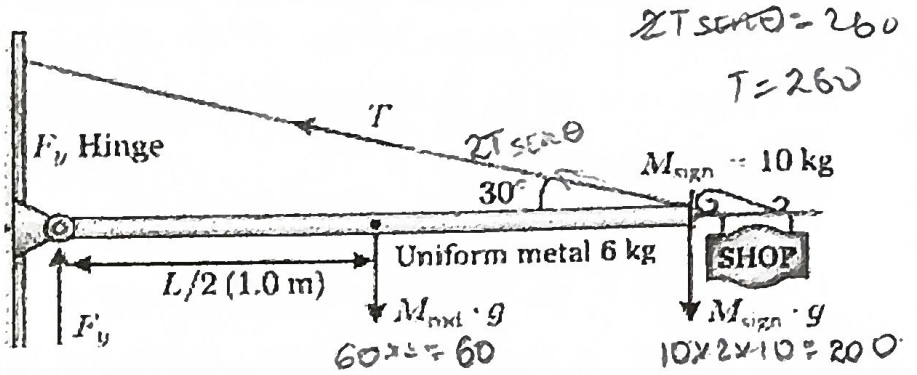
38. How much heat energy is required to raise the temperature of a 2 kg aluminum block by 20°C? (The specific heat capacity of aluminum is 900 J/kg·°C)

A. 18,000 J
B. 9,000 J
C. 36,000 J
D. 45,000 J

39. Two parallel metal plates are separated by a distance of 0.05 m. If a uniform electric field of 4,000 V/m exists between them, what is the potential difference between the plates?

A. 80 V B. 80,000 V C. 200 V D. 4,000 V

40. A uniform horizontal metal rod of length 2 meters and mass 6 kg is attached to a vertical wall via a frictionless hinge at its left end. The right end of the rod is supported by a thin steel cable that forms an angle of exactly 30^0 with the horizontal rod. A decorative shop sign of mass 10 kg hangs from the very end of the rod. What is the tension (T) in the supporting steel cable?



A. 130 N

B. 160 N

C. 200 N

D. 260 N

41. A capacitor stores a charge of 6×10^{-4} C when connected to a 12 V battery. What is the capacitance of this capacitor?

A. 5×10^{-5} F B. 2×10^{-5} F C. 2×10^{-4} F D. 7.2×10^{-3} F

42. A radioactive isotope sample has an initial activity rate of 800 Bq. If the half-life of this isotope is exactly 10 days, what will its activity be after 30 days have elapsed?

A. 400 Bq B. 100 Bq C. 200 Bq D. 266 Bq

43. Two long, straight, parallel conducting wires are separated by a distance d in a vacuum. Wire 1 carries a steady current I_1 pointing upward, and Wire 2 carries an identical steady current I_2 also pointing upward. Which of the following statements correctly describes the magnetic interaction forces between these two wires?

A. They exert an attractive force on each other, and doubling the separation distance d doubles the force.

B. They exert a repulsive force on each other, and doubling the separation distance d reduces the force to one-quarter.

C. They exert an attractive force on each other, and doubling the separation distance d reduces the force to one-half.

D. They exert a repulsive force on each other, and doubling the separation distance d reduces the force to one-half.

44. An object moves in a circular path of radius 2 m at a constant speed. If the centripetal acceleration of the object is measured to be 8 m/s^2, what is its linear speed?

A. 2 m/s B. 8 m/s C. 4 m/s D. 16 m/s

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45. A rigid flywheel begins rotating from rest with a constant angular acceleration of 3 rad/s^2. What will its total angular displacement be after rotating for 4 s?

A. 6 rad B. 48 rad C. 12 rad D. 24 rad

46. A uniform solid disk has a moment of inertia of 5 kg·m^2. If a constant net torque of 20 N·m is applied to the disk, what is its resulting angular acceleration?

A. 100.0 rad/s^2 B. 15.0 rad/s^2 C. 0.25 rad/s^2 D. 4.0 rad/s^2

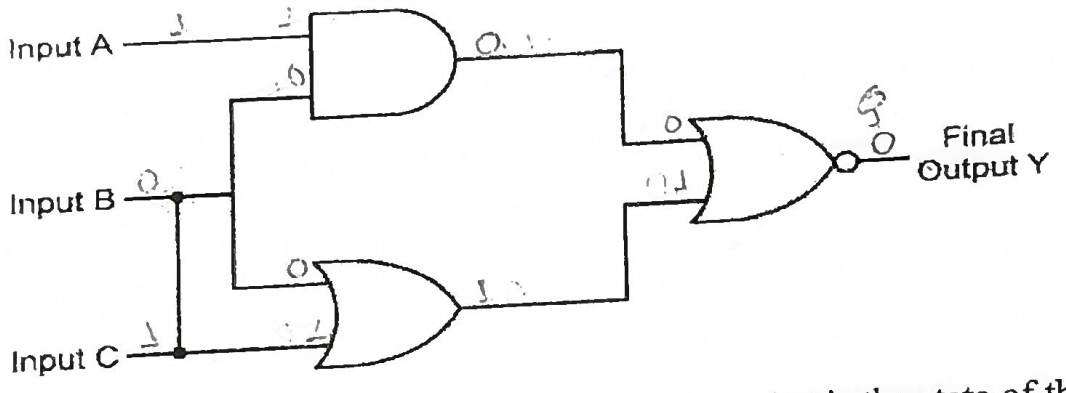
47. A solid metal object weighs 50 N in open air. When the object is completely submerged in water, its apparent weight drops to 38 N. What is the magnitude of the buoyant force exerted by the water on this object?

A. 88 N B. 12 N C. 50 N D. 38 N

48. Which of the following physical laws states that a changing magnetic flux through a conducting loop induces an electromotive force (emf) in that loop?

A. Ampere's Law B. Faraday's Law C. Ohm's Law D. Gauss's Law

49. Consider the logic circuit diagram shown below:



If the input states are set to A = 1, B = 0, and C = 1, what is the state of the final output Y?

A. 0 (Low) C. t will oscillate between 0 and 1

B. 1 (High) D. High-impedance open circuit

50. A transformer has 200 turns in its primary coil and 1,000 turns in its secondary coil. If an alternating input voltage of 120 V is applied to the primary coil, what is the output voltage across the secondary coil?

A. 1,200 V B. 600 V C. 240 V D. 24 V

51. An electrical engineer requires a solid-state circuit component that can act as an electronic switch or amplify weak alternating current signals. Which component should they select?

A. A variable carbon resistor C. A simple p-n junction diode

B. A bipolar junction transistor (BJT) D. A standard step-down transformer

52. A wooden crate of mass 10 kg is released from rest at the top of a rough inclined plane that makes an angle of 37^0 with the horizontal. The coefficient of kinetic friction between the crate and the incline is 0.25. What is the acceleration of the crate as it slides down the ramp?

A. 2.0 m/s^2 B. 4.0 m/s^2 C. 6.0 m/s^2 D. 8.0 m/s^2

53. Kirchhoff's Junction Rule (Current Law) and Kirchhoff's Loop Rule (Voltage Law) are fundamental tools used to analyze electrical networks. These two rules are direct consequences of which physical conservation laws, respectively?

A. Conservation of Momentum and Conservation of Charge

B. Conservation of Energy and Conservation of Mass

C. Conservation of Charge and Conservation of Energy

D. Conservation of Mass and Conservation of Momentum

54. A heavy stone block is placed on an adjustable inclined plane. A lab technician slowly raises the angle of the incline. The block remains completely stationary until the ramp reaches an angle of exactly 45^0 with the horizontal, at which point the block just begins to budge and slide downward. What is the coefficient of static friction between the block and the incline surface?

A. 1.00 B. 0.71 C. 0.50 D. 0.25

55. Consider a logic circuit with two inputs, A and B. The inputs are first passed through individual NOT gates, and their inverted outputs ($\bar{A}$ and $\bar{B}$) are then fed directly into a single standard OR gate. This entire circuit combination behaves identically to which single logic gate?

A. AND B. NAND C. NOR D. XOR

56. A flat circular loop of wire consisting of exactly 100 turns and enclosing a cross-sectional area of 0.05 m^2 is placed perpendicular to a uniform magnetic field. The magnetic field drops uniformly from an initial strength of 0.8 T down to 0.2 T over a time interval of exactly 3 seconds. What is the magnitude of the average electromotive force (emf) induced in the loop?

A. 0.01 V B. 0.10 V C. 1.00 V D. 10.00 V

57. A beam of monochromatic light travels through an analytical liquid and approaches a flat boundary with an open-air environment ($n_{air} = 1.00$). If the index of refraction of the liquid is exactly $n = 2.00$, what is the critical angle (θ) above which total internal reflection will occur?

A. 15^0 B. 30^0 C. 45^0 D. 60^0

58. Moving a test charge of 2.5 x 10^{-6} C from point A to point B within a static electric field requires exactly 6.0 x 10^{-4} J of work. What is the potential difference ($V_B - V_A$) between these two points?

A. 0.0015 V B. 4.1 X 10^{-10} V C. 240 V D. 1500 V

59. A specific copper wire has an electrical resistance R. A second copper wire (Wire B) is manufactured such that it has the exact same total length as the first wire, but its cross-sectional diameter is exactly 0.5 times the diameter of the first wire. What is the electrical resistance of Wire B?

A. 0.5 R B. 2 R C. 4 R D. 16 R

60. A typical 220-V mains power line in an Ethiopian residential home is protected by a 15-A safety fuse. What is the maximum number of standard 220-V, 500 W light bulbs that can be switched on and operated simultaneously at full brightness without blowing the fuse?

A. 4 B. 6 C. 8 D. 12

The End

BLESSED G. MICHAEL CATHOLIC SECONDARY SCHOOL

MODEL-EXAM Answer Sheet **Subject:** Physics .

Allotted time: 2:10hr

Name Hanna Ayalew **Sec.** 12TVC **No.** 21

58

Direction: Blacken the circles given next to the numbers as per your answer's choice.

	A	B	C	D
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114.	O	O	O	O
115.	O	O	O	O

Hanna Ayalew

BAHIR DAR CITY ADMINISTRATION EDUCATION DEPARTMENT PHYSICS MODEL EXAMINATION FOR GRADE 12 IN 2018 E.C

TIME ALLOWED; 2:30hrs

Use: $g=10m/s^2$, $\rho_{water}=1000kg/m^3$, $\rho_{air}=1.29kg/m^3$, $\mu_0=4\pi\times10^{-7}Tm/A$, $P_{atm}=1\times10^5N/m^2$, $\sin30^0=\cos60^0=0.5$ and $\cos30^0=\sin60^0=0.87$, when necessary.

I- THIS BOOKLET CONTINS 50 MULTIPLE CHOICE PHYSICS QUESTIONS. THERE ARE ALSO FOUR ALTERNATIVES TO EACH QUESTIONS. CHOOSE THE MOST APPROPRIATE ANSWER AMONG THE GIVEN ALERNATIVES AND WRITE THE LETTER OF YOUR CHOICE ON YOUR ANSWER SHEET.

C 1. Which one of the following is true about rough surface inclined plane in simple machine?

A. The velocity ratio is different.

B. The actual mechanical advantage is constant.

C. The actual mechanical advantage depends on the roughness of the surface.

D. The efficiency of an inclined plane is similar for rough & smooth surfaces.

B 2. By what factor will the intensity of sound wave changes as the distance from the source is doubled?

A. It is decreased by a factor of 1/2

B. It's decreased by a factor of 1/4

C. It's decreased by a factor by a factor of 2

D. It's decreased by a factor of 4

A 3. what is the relationship between tensile stress and tensile strain for most solid materials?

A. up to elastic limit tensile stress is directly proportional to tensile strain.

B. Up to elastic limit tensile stress is inversely proportional to tensile strain.

C. The product of tensile stress and tensile strain is constant.

D. there is no any relationship between tensile stress and tensile strain.

B 4. Among the following one is not the purpose of simple machine?

A. simple machine can multiply force.

B. simple machine can multiply energy.

C. simple machine can multiply speed.

D. simple machine changes direction of motion

D 5. Sound wave is a longitudinal mechanical wave that travels in particle media with different speeds. Which of the following statement is correct about its speed?

A. sound travels faster in liquids than solids.

B. Sound travels faster in air than liquids.

C. sound travels faster in air than solids.

D. Sound travels faster in solids than liquids & air.

A 6. What will be the approximate mass of air in a living room of dimensions [4mx5mx3m]?

A. 58.05kg

B. 58g

C. 1kg

D. 100g

C 7. When the effort is in between the load and the fulcrum, the order of the lever will be:

A. first class

B. second class

C. third class

D. force multiplier

8. which of the following is true about an image formed by convex mirror?

A. virtual, erect and enlarged
B. real, inverted and enlarged
C. real, diminished and inverted
D. virtual, erect and diminished

B 9. A concave lens has a focal length of 15cm. at what distance from the lens should an object be placed ,so that it forms an image at 10cm from the lens?

A. 20cm
B. 30cm
C. 10cm
D. 0.1cm

A 10. The defect of myopia /near sightedness/ can be corrected by using eye glass made from:

A. concave lens of suitable power of magnification
B. convex lens of suitable magnification
C. concave mirror of suitable magnification
D. convex mirror of suitable magnification

A 11. When a force (F) is applied on an object of mass(m) 4kg,the object moves from 5m/s to a final velocity(v), what will be the final velocity if the work done is decreased to 32J ?

A. 3m/s
B. 9m/s
C. 16m/s
D. 25m/s

A 12. When a block of mass(m) 50kg is moving freely downward on a smooth inclined plane with 30^0 inclination from the horizontal as shown in the figure below: what will be will be the acceleration of the block?

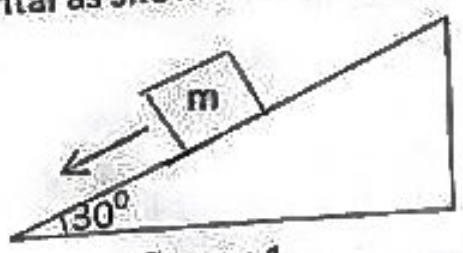


figure: 1

A. $5m/s^2$
B. $10m/s^2$
C. $20m/s^2$
D. $15m/s^2$

A 13. Two spheres of identical mass (m)are collide on a smooth horizontal surface, if m_1 moves to the right with an initial speed of u_1 and m_2 is at rest, what will be their velocity after collision if they move stick together?

A. $1/2u_1$
B. $2u_1$
C. $1/4u_1$
D. $4u_1$

A 14. A hydraulic lift has a small piston with a surface area of $0.002m^2$ and a larger piston with surface area of $0.2m^2$. Assume your mass is 60kg. If you stand on the smaller piston, how much mass can you lift at the larger piston?

A. 6oookg
B. 60000kg
C. 60kg
D. 0.6kg

C 15. A body accelerates uniformly from rest to a speed of 30m/s in 5 seconds. What will be the distance travelled by the body?

A. 80m
B. 150m
C. 75m
D. 6m

D 16. Let vector A=3units and vector B= 4units. Which one of the following **cannot** be the possible value of the magnitude of their resultant?

A 7unit
B. 1unit
C. 5unit
D. 8unit

B 17. A long straight current carrying wire produces a magnetic field of 0.8T at a distance of 2cm from the wire. What is the magnetic field strength at a distance of 4cm from the wire?

A. 1.6T
B. 0.4T
C. 4T
D. 2T



18. Which of the following is not the main objective about the purpose of establishing physics community?

A. Enhancing physics research and education
B. Organizing physics conferences
C. Promoting active participants in physics
D. Avoid using indigenous knowledge

19. Which one of the following is correct?

A. Temperature and heat are two names of the same thing
B. temperature measures motion of particles of an atom and molecules
C. temperature and heat have the same SI unit
D. heat is average kinetic energy and temperature is the total energy transferred

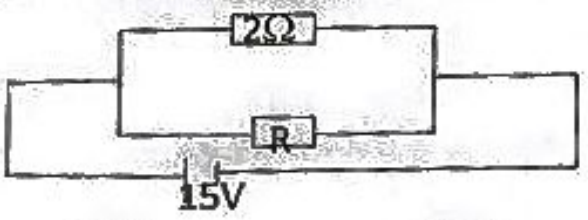
20. Among the following one is **not true** for a conductor in electrostatic equilibrium

A. the interior electric field is not zero
B. the exterior electric field is perpendicular to the surface
C. the surface is an equipotential
D. The interior is at the same potential as the surface

21. The equivalent resistance value for a group of parallel resistors is:

A. less than any individual resistor in the group
B. equal to the smallest resistor in the group
C. equal to the average resistance of the group
D. larger than any resistor in the group

22. for the following electric schematic circuit, the total voltage and power are 15V and 150w respectively, what will be the value for resistor 'R'?



A. 9Ω B. 6Ω C. 20Ω D. 10Ω

23. As the dielectric material is inserted between the parallel plate capacitor, which of the following statement is **not** correct?

A. capacity of the capacitor is increases
B. the electric field increases,
C. the energy stored in the capacitor increases
D. The voltage increases

24. Which one of the following is the working principle of radar?

A. the echo principle
B. the principle of total internal reflection
C. The principle of refraction of wave
D. the principle of interference of a wave

25. In an inclined projectile motion what is the relationship between range(R) and maximum height(H)?

A. H=4tanθ B. R= H tanθ/4 C. H= R tanθ/4 D. H=R

26. Semiconductor with small impurities of trivalent or pentavalent atom is:
A. Intrinsic semiconductor B. Extrinsic semiconductor
C. Impurity acceptor or donor D. All except A

27. A ball is thrown horizontally with the velocity of 30m/s from the top of 80m building. What is the speed of the ball at the instant before it strikes the ground at the bottom of the building?
A. 40m/s B. 30m/s C. 50m/s D. 100m/s

28. For a transistor, the emitter-base junction will be connected:
A. forward biased B. reverse biased
C. Zero biased D. rectified

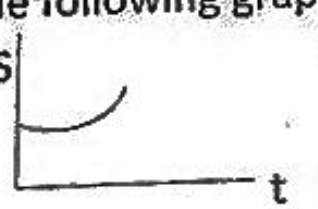
29. which one of the following is true about Keppler's 3rd law, if 'T' is the period of a planet and 'R' is the mean distance between the sun and the planet?
A. T^2R^3 is constant B. T^2R^{-3} is constant C. T^2R^{-3} is constant D. T~R

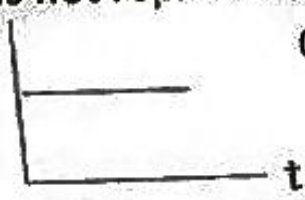
30. Two masses are attracted to each other by a gravitational force (F). if the distance between the masses is halved, what is the new gravitational force between them in terms of 'F'?
A. 4F B. 2F C. 1/2F D. 1/4F

31. one of the following cannot be the unit of pressure?
A. mmHg B. Torr C. Psi D. N/m

32. Lenz's law represents :
A. The relation between current and magnetic field
B. The relation between magnetic force and magnetic field
C. The relation between induced electromotive force (emf) and rate of change of flux
D. The relation between electromotive force and electric field.

33. Which one of the following uses the application of Pascal's principle?
A. Newton-meter B. Hydraulic press
C. Lifting up of an airplane D. motion of rocket

34. Which of the following graph does not represent uniformly accelerated motion in a straight line?
A. 

B. 

C. 

D. 

35. which one of the following is true about steady fluid flow?
A. A velocity of every fluid particle arriving at a given point cannot vary with time.
B. The arrival of each fluid particle at a given point varies with time.
C. The velocity of each fluid particle to reach at a given point varies with time.
D. The motion of the fluid is not predictable

36. The specific gravity of aluminum(Al) is 2.7. What is the mass of aluminum with volume of $100cm^3$?
A. 270g B. 2700g C. 2.7g D. 27g

37. The square of the resultant of two equal vectors is equal to 4 times their dot product. W

will be the angle between the two vectors?

A. 180^0 B. 90^0 C. 0^0 D. 45^0

A 38. An ideal transformer with 150 watt has an input voltage of 10volts and an output current of 5A. What is the ratio of output voltage on the secondary coil to the input voltage on the primary coil?

A. 3:1 B. 1:3 C. 1:5 D. 1:2

C 39. Two objects 'A'&'B' have the same volume, but 'A' is larger than 'B' and they are completely submerged in oil . Which one of the following is true about the buoyant force exerted by the oil on the two objects?

A. The buoyant force on 'B' is greater than the buoyant force on 'A'.

B. The buoyant force on 'A' is greater than the buoyant force on 'B'

C. The buoyant forces on 'A' and 'B' are equal.

D. There is no enough information to compare the buoyant forces on objects 'A' and 'B'

A 40. Among the given alternatives one is **not** the source of magnetic field?

A. stationary charged particles B. moving charged particles

C. current carrying conductors D. lode stone

C 41. Continuity of intrinsic semiconductor will be:

A. Decrease with rise in temperature B. Increase with decrease in temperature

C. Increase with rise in temperature D. Decrease with decrease in temperature

A 42. The absolute pressure below the surface of fresh water is three times the atmospheric pressure. What is the depth of this fresh water?

A. 20m B. 10m C. 5m D. 30m

B 43. The magnetic field of a 5A current carrying wire at the radius of 2cm from the wire is :

A. $0.5\mu_0 T$ B. $50\mu_0 T$ C. $0.25\mu_0 T$ D. $0.05\mu_0 T$

C 44. An electrical machine which is static device that works in an alternative current system and used to change the voltage level is:

A. motor B. generator

C. transformer D. Relay/contactor

C 45. An electronic semiconductor which coverts electric signals from Ac to DC is :

A. oscillator B. transformer C. diode D. capacitor

C 46. which one of the following is **not true** about projectile motion?

A. There is no acceleration in the horizontal component of the projectile motion.

B. There is constant downward free fall acceleration.

C. At the maximum height the speed of the projectile is zero.

D. The velocity vector of the projectile is always tangent to the trajectory.

B 47. The change in magnetic flux linked with a coil is given by $\Delta\Phi_B$=18weber. The induced emf in the coil at Δt=0.75 sec is:

A. 20V B. 24V C. 30 V D. 54V



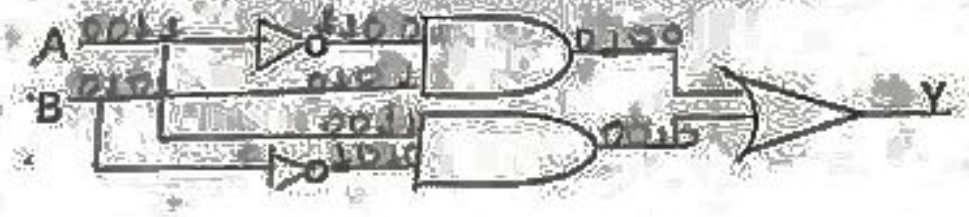
48. A car accelerates from 20m/s to 24m/s in 5 seconds. What will be the angular acceleration of the wheels of the car if the radius of the wheel is 40cm?

A. 0.5rad/s^2 B. 2rad/s^2 C. 0.32rad/s^2 D. 10rad/s^2

49. Which one of the following is **not** true about magnetic field lines?

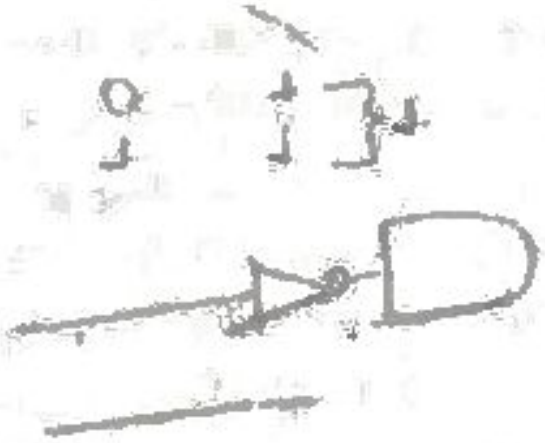
A. They have known starting and ending points.
B. They form closed loop.
C. They are denser near the poles
D. They never intersect with each other

50. Given the following logic circuit diagram. Which alternative value is the correct value of the output 'Y' in the truth table?



A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

A. 0,1,0,1 B. 0,1,1,0
C. 1,0,0,1 D.1,0,1,0





BLESSED G. MICHAEL CATHOLIC SECONDARY SCHOOL

MODEL-EXAM ONE Answer Sheet **Subject:** Physics

Allotted time: 1:40'

Name Hanna Ayalew **Sec.** 12th C **No.** 21

49

Direction: Blacken the circles given next to the numbers as per your answer's choice.

	A	B	C	D
1.	O	O	●	O
2.	O	●	O	O
3.	●	O	O	O
4.	O	●	O	O
5.	O	O	O	●
6.	●	O	O	O
7.	O	O	●	O
8.	O	O	O	●
9.	O	●	O	O
10.	●	O	O	O
11.	●	O	O	O
12.	●	O	O	O
13.	●	O	O	O
14.	●	O	O	O
15.	O	O	●	O
16.	O	O	O	●
17.	O	●	O	O
18.	O	O	O	●
19.	O	●	O	O
20.	●	O	O	O
21.	●	O	O	O
22.	O	●	O	O
23.	O	O	O	●
24.	●	O	O	●
25.	O	O	●	O
26.	O	●	O	O
27.	O	O	●	O
28.	●	O	O	O
29.	O	●	O	O
30.	●	O	O	O

	A	B	C	D
31.	O	O	O	●
32.	O	O	●	O
33.	O	●	O	O
34.	O	O	O	●
35.	●	O	O	O
36.	●	O	O	O
37.	O	O	●	O
38.	●	O	O	O
39.	O	O	●	O
40.	●	O	O	O
41.	O	O	●	O
42.	●	O	O	●
43.	O	●	O	O
44.	O	O	●	O
45.	O	O	●	O
46.	O	O	●	O
47.	O	●	O	O
48.	O	●	O	O
49.	●	O	O	O
50.	O	●	O	O
51.	O	O	O	O
52.	O	O	O	O
53.	O	O	O	O
54.	O	O	O	O
55.	O	O	O	O
56.	O	O	O	O
57.	O	O	O	O
58.	O	O	O	O
59.	O	O	O	O
60.	O	O	O	O

	A	B	C	D
61.	O	O	O	O
62.	O	O	O	O
63.	O	O	O	O
64.	O	O	O	O
65.	O	O	O	O
66.	O	O	O	O
67.	O	O	O	O
68.	O	O	O	O
69.	O	O	O	O
70.	O	O	O	O
71.	O	O	O	O
72.	O	O	O	O
73.	O	O	O	O
74.	O	O	O	O
75.	O	O	O	O
76.	O	O	O	O
77.	O	O	O	O
78.	O	O	O	O
79.	O	O	O	O
80.	O	O	O	O
81.	O	O	O	O
82.	O	O	O	O
83.	O	O	O	O
84.	O	O	O	O
85.	O	O	O	O
86.	O	O	O	O
87.	O	O	O	O
88.	O	O	O	O
89.	O	O	O	O
90.	O	O	O	O

	A	B	C	D
91.	O	O	O	O
92.	O	O	O	O
93.	O	O	O	O
94.	O	O	O	O
95.	O	O	O	O
96.	O	O	O	O
97.	O	O	O	O
98.	O	O	O	O
99.	O	O	O	O
100.	O	O	O	O
101.	O	O	O	O
102.	O	O	O	O
103.	O	O	O	O
104.	O	O	O	O
105.	O	O	O	O
106.	O	O	O	O
107.	O	O	O	O
108.	O	O	O	
109.	O	O	O	
110.	O	O	O	

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G-12 PHYSICS Model Exam I **Ginbot 2018 E.C** **Time: 1:30**

Name: Hanna Ayalew **Gr-& sec:** _____ **Number:** _____

For each of the following questions, choose the best answer from the given alternatives.

B 1. A ball is kicked at an angle of 30° to the horizontal. If the horizontal component of its velocity is 17.3 m/s, what is the initial speed of the ball?

A. 10 m/s B. 20 m/s C. 30 m/s D. 34.6 m/s

C 2. In a hydraulic press, a force of 50 N is applied to a small piston of area 0.01 m^2. If the large piston has an area of 0.1 m^2, what is the maximum weight it can lift?

A. 5 N B. 50 N C. 500 N D. 5000 N

B 3. Which of the following describes the relationship between physics and engineering?

A. Physics is the practical application of engineering.

B. Engineering uses physical laws to design and build technology.

C. They are completely unrelated fields.

D. Physics only deals with theories, while engineering only deals with tools.

A 4. A step-down transformer reduces 240 V to 12 V. If the primary coil has 4000 turns, how many turns are in the secondary coil?

A. 200 B. 800 C. 2000 D. 20

C 5. In a p-type semiconductor, the majority charge carriers are:

A. Electrons B. Neutrons C. Holes D. Protons

C 6. If the displacement of an object is given by ($s = 5t + 2t^2$), what is the initial velocity?

A. 0 m/s B. 2 m/s C. 5 m/s D. 9 m/s

B 7. Vector **A** has a magnitude of 6 units and points North. Vector **B** has a magnitude of 8 units and points East. The magnitude of the resultant vector $\mathbf{R} = \mathbf{A} + \mathbf{B}$ is:

A. 2 units B. 10 units C. 14 units D. 48 units

C 8. An object is moving in a circle of radius (r) with constant speed (v). If the speed is doubled and the radius is halved, the centripetal acceleration becomes:

A. 2 times larger B. 4 times larger C. 8 times larger D. Remains the same

C 9. A 2 kg block is pushed with a force of 10 N on a horizontal surface where the friction force is 2 N. The acceleration of the block is:

A. 5 m/s^2 B. 6 m/s^2 C. 4 m/s^2 D. 12 m/s^2

B 10. A body is thrown vertically up with 20 m/s. How long does it take to reach the top?

A. 1 s B. 2 s C. 4 s D. 10 s

$\cos 30° = \sin 60° = \frac{\sqrt{3}}{2}, \cos 37° = \sin 53° = 0.8, \cos 45° = \sin 45° = \frac{\sqrt{2}}{2}, \cos 53° = \sin 37° = 0.6, \cos 60° = \sin 30° = 0.5$

($g = 10\ m/s^2$ $k = 9 \times 10^9\ Nm^2/C^2$ $c = 3 \times 10^8\ m/s$)

11. How much heat is required to raise the temperature of 2 kg of water from 20°C to 30°C? ($c = 4200$ J/kg°C)

A. 42,000 J B. 84,000 J C. 8,400 J D. 4,200 J

12. Two parallel wires carrying current in the same direction will:

A. Repel each other C. Rotate each other

B. Attract each other D. Have no effect on each other

13. In a radioactive decay, a nucleus emits a Beta particle. The atomic number of the daughter nucleus:

A. Increases by 1 C. Decreases by 2

B. Decreases by 1 D. Remains the same

14. A car accelerates from 10 m/s to 30 m/s over a distance of 100 m. Its acceleration is:

A. 2 m/s^2 B. 4 m/s^2 C. 8 m/s^2 D. 1 m/s^2

15. A uniform meter rule is balanced at the 50 cm mark. If a 2 N weight is hung at the 10 cm mark, where must a 4 N weight be hung to maintain equilibrium?

A. 20 cm mark B. 70 cm mark C. 80 cm mark D. 90 cm mark

16. The resistance of a wire is (R). If the length is doubled and the area is doubled, the new resistance is:

A. (R/2) B. (R) C. (2R) D. (4R)

17. An object is placed 15 cm in front of a converging lens with a focal length of 10 cm. The image produced is:

A. Real, inverted, and magnified C. Virtual, upright, and magnified

B. Real, inverted, and diminished D. Virtual, upright, and diminished

18. A machine has a Velocity Ratio (VR) of 5 and an efficiency of 80%. What is its Mechanical Advantage (MA)?

A. 4 B. 6.25 C. 0.16 D. 400

19. A sound wave has a frequency of 500 Hz and a wavelength of 0.7 m. The speed of sound is:

A. 340 m/s B. 350 m/s C. 714 m/s D. 1500 m/s

20. A 60 W bulb is left on for 10 hours. How much energy in Joules is consumed?

A. 600 J B. 36,000 J C. 2,160,000 J D. 60 J

21. An object falls freely from rest. The distance it covers in 3 seconds is approximately

A. 30 m B. 45 m C. 90 m D. 15 m

22. Geophysics is most useful in which of the following activities?

A. Predicting chemical reactions C. Designing aircraft wings

B. Searching for underground oil and minerals D. Studying plant growth

C 23. The half-life of a substance is 10 days. If you start with 80 g, how much is left after 30 days?
A. 40 g B. 20 g C. 10 g D. 5 g

B 24. According to Bernoulli's Principle, in a pipe where the diameter decreases:
A. Velocity increases and pressure increases
B. Velocity increases and pressure decreases
C. Velocity decreases and pressure increases
D. Velocity remains constant

C 25. A tennis ball of mass 0.10 kg traveling horizontally at 40.0 m/s is struck back by a racket, returning in the opposite direction at 30.0 m/s. What is the magnitude of the impulse delivered to the ball?
A. 1.0 Ns B. 4.0 Ns C. 7.0 Ns D. 12.0 Ns

B 26. In a longitudinal wave, the particles of the medium move:
A. Perpendicular to the wave direction
B. Parallel to the wave direction
C. In circles
D. They do not move

B 27. A thermometer uses the property of thermal expansion. This is a:
A. Chemical property
B. Thermometric property
C. Electrical property
D. Nuclear property

28. For a rigid body to be in static equilibrium, the sum of all forces must be zero and:
A. The object must be moving
B. The sum of all torques must be zero
C. The velocity must be changing
D. The mass must be zero

C 29. Which electromagnetic wave is used in TV remote controls?
A. Ultraviolet B. X-rays C. Infrared D. Gamma rays

C 30. A real image formed by a convex lens is always:
A. Upright B. Virtual C. Inverted D. Smaller than the object

A 31. If the net work done on an object is positive, the kinetic energy:
A. Increases B. Decreases C. Remains constant D. Becomes zero

D 32. The SI unit of pressure, the Pascal, is equivalent to:
A. N/m B. Nm C. Nm^2 D. kg/ms^2

A 33. Specific heat capacity is the heat required to raise the temperature of:
A. 1 kg of a substance by 1°C
B. 10 kg of a substance by 1°C
C. The whole body by 1°C
D. 1 kg of a substance to its boiling point

A 34. An object is moving in a circle. If its angular velocity increases from 2 rad/s to 6 rad/s in 2 seconds, what is the angular acceleration?
A. 2 rad/s^2 B. 4 rad/s^2 C. 8 rad/s^2 D. 1 rad/s^2

Set by: Abebaw Tilahun (with AI)

Contact: +251 921 279 547 or @abty01

($g = 10\ m/s^2$ $k = 9 \times 10^9\ Nm^2/C^2$ $c = 3 \times 10^8\ m/s$)

35. An electric motor converts:
A. Mechanical energy to electrical
B. Electrical energy to mechanical
C. Chemical energy to electrical
D. Heat energy to mechanical

36. In an NPN transistor, the arrow on the emitter points:
A. Toward the base
B. Away from the base
C. Toward the collector
D. It has no arrow

37. The weight of an object is:
A. The same as its mass
B. The gravitational force acting on it
C. Measured in kilograms
D. Independent of location

38. A hydraulic system is an application of:
A. Archimedes' Principle
B. Newton's Third Law
C. Pascal's Law
D. Hooke's Law

39. If a wire is moved through a magnetic field, an e.m.f. is induced. This is:
A. Self-induction
B. Electromagnetic induction
C. Mutual induction
D. Magnetic resonance

40. Nuclear fusion occurs when:
A. A heavy nucleus splits
B. Light nuclei combine
C. Electrons are emitted
D. Neutrons are absorbed

41. The focal length of a concave mirror is 10 cm. If an object is placed 20 cm from the mirror, the image is:
A. Virtual and upright
B. Real and inverted at the same size
C. Real and magnified
D. Real and diminished

42. A simple machine has an efficiency of 80%. If the work input is 200 J, the useful work output is:
A. 250 J B. 160 J C. 120 J D. 80 J

43. The speed of sound in air increases when:
A. Humidity decreases
B. Temperature increases
C. Frequency increases
D. Pressure decreases

44. Magnetic flux is maximum when the angle between the magnetic field and the area vector is:
A. 0° B. 90° C. 180° D. 45°

45. A logic gate that acts as an "inverter" is the:
A. AND gate B. OR gate C. NOT gate D. XOR gate

46. Refraction occurs because light changes its _____ when moving between media.
A. Color B. Frequency C. Speed D. Phase

47. Two copper wires have the same length, but wire A is twice as thick as wire B. The resistance of wire A is:

A. Twice that of B
B. Half that of B
C. One-fourth that of B
D. Four times that of B

48. According to Lenz's Law, the induced current in a circuit always:

A. Opposes the change that produced it
B. Supports the change that produced it
C. Flows from North to South
D. Increases the magnetic flux

49. Which logic gate produces a 'HIGH' output only when both inputs are 'HIGH'?

A. OR B. NOT C. AND D. NAND

50. In a projectile motion, the maximum range is achieved at an angle of:

A. 30° B. 45° C. 60° D. 90°

51. In fluid flow, where the velocity of the fluid is high, the pressure is:

A. High B. Low C. Zero D. Atmospheric

52. A permanent magnet attracts:

A. All metals
B. Ferromagnetic materials
C. Plastic
D. Aluminum only

53. In a transformer, the primary coil has 200 turns and the secondary has 800 turns. If the input AC voltage is 12V, the output voltage is:

A. 3V B. 12V C. 48V D. 96V

54. If the magnitude of the resultant of two equal vectors is equal to the magnitude of each vector, the angle between them is:

A. 0° B. 60° C. 90° D. 120°

55. A car traveling at 20 m/s brakes to a stop in 5 seconds. What is its displacement during this time?

A. 100 m B. 50 m C. 250 m D. 10 m

56. An example of a second-class lever is:

A. Scissors B. Wheelbarrow C. Tweezers D. Seesaw

57. Human hearing range is approximately:

A. 2 Hz - 200 Hz
B. 20 Hz - 20,000 Hz
C. 20,000 Hz - 2 MHz
D. 0 Hz - 10 Hz

58. Three resistors of (6Ω) each are connected in parallel. Their equivalent resistance is:

A. 18 Ω B. 6 Ω C. 2 Ω D. 0.5 Ω

($g = 10\ m/s^2$ $k = 9 \times 10^9\ Nm^2/C^2$ $c = 3 \times 10^8\ m/s$)

59. When 100J of heat is added to a system and the system does 40J of work, what is the change in internal energy? $-40 + 100 = +60$

A. 140 J B. 60 J C. -60 J D. 40 J

60. The current gain (β) of a transistor is 100. If the base current $I_b = 50\mu A$, then what is the collector current I_c? $100 = \frac{I_c}{50 \times 10^{-6}}$ $I_c = 50 \times 10^{-4} = 5 \times 10^{-3} = 5$

A. 5 mA B. 50 mA C. 0.5 mA D. 500 mA

60/60 Smooth

G-12 PHYSICS Model Exam I **Ginbot 2016 E.C** **Time: 1:30**

Name: Hanna Ayalew **Gr-& sec:** 12th C **Number:** 24

1.	B	16.	B	31.	A	46.	C
2.	C	17.	A	32.	D	47.	C
3.	B	18.	A	33.	A	48.	A
4.	A	19.	B	34.	A	49.	C
5.	C	20.	E A	35.	B	50.	B
6.	C	21.	B	36.	B	51.	B
7.	B	22.	B	37.	B	52.	B
8.	C	23.	C	38.	C	53.	C
9.	C	24.	B	39.	B	54.	D
10.	B	25.	C	40.	B	55.	B
11.	B	26.	B	41.	B	56.	B
12.	B	27.	B	42.	B	57.	B
13.	A	28.	B	43.	B	58.	C
14.	AB	29.	C	44.	A	59.	B
15.	B	30.	C	45.	C	60.	A